Computational Mathematics and Statistics

End Term Examination Solutions (E1PA101T)

Institution: Galgotias University
Session: Jan 2026
Program: Masters of Computer Applications (MCA)

1. Matrix Operations

Calculate the value of **10C - 3D + 5E** given matrices C, D, and E.

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C = [ 1  2 ]   D = [ 11  4 ]   E = [ 1  4 ]
    [ 7  1 ]       [  5  8 ]       [ 5  1 ]

10C = [ 10  20 ]
      [ 70  10 ]

-3D = [ -33 -12 ]
      [ -15 -24 ]

5E  = [  5  20 ]
      [ 25   5 ]

Result = 10C - 3D + 5E
       = [ (10 - 33 + 5)   (20 - 12 + 20) ]
       = [ (70 - 15 + 25)  (10 - 24 + 5)  ]

Final Matrix = [ -18  28 ]
               [  80  -9 ]
```

2. Standard Deviation

Series: 3, 6, 8, 10, 12, 14

```
Mean (x̄) = (3+6+8+10+12+14) / 6 = 53 / 6 ≈ 8.83
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Deviations (x - x̄)²:
(3 - 8.83)² = 34.03
(6 - 8.83)² = 8.01
(8 - 8.83)² = 0.69
(10 - 8.83)² = 1.37
(12 - 8.83)² = 10.05
(14 - 8.83)² = 26.73

Sum of Squares = 80.88
Variance = 80.88 / 6 = 13.48
Standard Deviation = √13.48 ≈ 3.67
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3. Conditional Probability

Definition: The probability of an event occurring given that another event has already occurred.
Problem: P(Science) = 4/5, P(Science ∩ Maths) = 1/2. Find P(Maths | Science).

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Formula: P(B|A) = P(A ∩ B) / P(A)
P(Maths | Science) = (1/2) / (4/5)
                   = (1/2) * (5/4)
                   = 5/8 (or 0.625)
```

4. Central Limit Theorem (CLT)

Theorem: As sample size increases, the distribution of sample means approaches a normal distribution, regardless of the population's distribution.
Problem: $\mu = 1000$, $\sigma = 100$, $n = 36$. Find $P(980 < \bar{x} < 1020)$.

```
Standard Error (SE) = σ / √n = 100 / √36 = 16.67
Z-scores for 980 and 1020:
Z = (980 - 1000) / 16.67 = -1.2
Z = (1020 - 1000) / 16.67 = 1.2
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P(-1.2 < Z < 1.2) = 0.884 (as given)
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5. Discrete Probability Distributions

Find k for f(x) = k(x^3 + 1) for x = 1, 2, 3, 4.

```
Sum of f(x) = 1
k[(1³+1) + (2³+1) + (3³+1) + (4³+1)] = 1
k[2 + 9 + 28 + 65] = 1
k[104] = 1
k = 1/104

Mean (E[X]) = Σ x * f(x) = (1*2 + 2*9 + 3*28 + 4*65) / 104
            = (2 + 18 + 84 + 260) / 104 = 364 / 104 ≈ 3.5
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